

Prachi Sanaliya
AU2340035

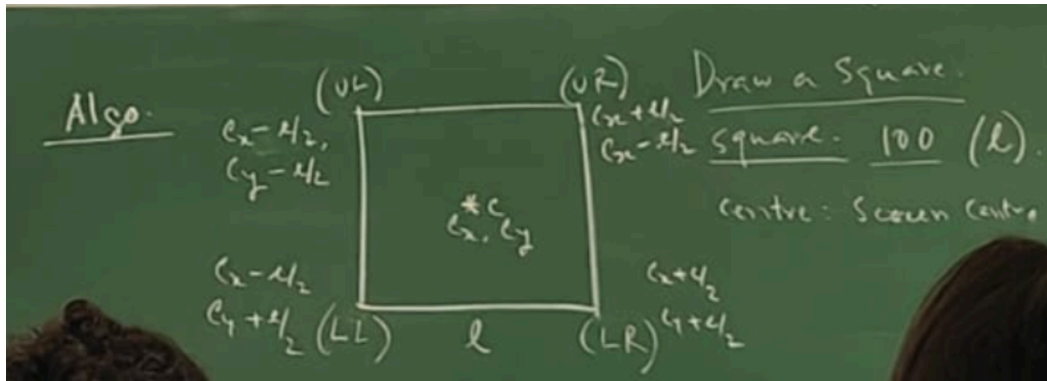
CSC_360
Class Notes

18/8/26

- Screen coordinate system and dimension of pixels in Computer Graphics.
- Creating geometric square using the centre of the screen as the reference point.
- Understanding the way in which the same pixel dimension appears different on screens with varying physical dimensions.
- Understanding the basic structure of Maven Project.
- Introduction to javax.swing and java.awt for Java graphics programming.
- Understanding JFrame, setColor() and graphical drawing.
- Introduction to object field and static field in Java.

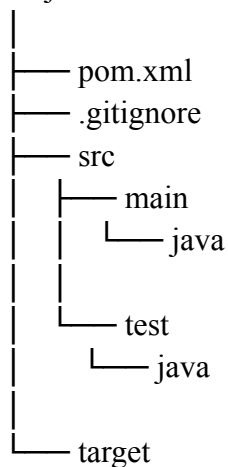
- Screen Coordinate System and Pixel Dimension:
 - ~ In talking about dimensions of a square, there is no need to specify any unit by default.
 - ~ Dimension of shapes in computer graphics is usually in pixels.
 - ~ For instance, a square of 100 means a square of 100 x 100 pixels.
 - ~ Pixel dimension of the square can appear different on different sized physical screens.
 - ~ This is due to differences arising from such considerations as screen resolution and pixel density.Hence, pixel dimensions and physical dimension of the screen are not always the same.

- Using Centre of the Screen for Drawing:
 - ~ The centre of the screen can be used as the reference point for drawing.
 - ~ It is not necessary to specify an arbitrary reference point since centre is desired as the reference point.



- The simple Maven project structure consists of the following files:

Project



- Important Elements:
 - ~ pom.xml – Maven configuration file.
 - ~ src/main/java – main Java sources.
 - ~ src/test/java – testing sources.
 - ~ target/ – contains generated files during the build procedure.
 - ~ .gitignore – defines which files/directories must not be managed by Git.
 - ~ target/ is usually included in .gitignore since it stores generated build files.

- Java graphics libraries:

javax.swing

- ~ Provides graphical user interface and window creation utilities.
- ~ JFrame creates the main application window.
- ~ JPanel provides drawing area or canvas.

Java.awt

- ~ Contains all necessary classes and tools for graphics.
- ~ Graphics and Graphics2D classes provide means to draw graphical objects.
- ~ Color class specifies colors.
- ~ BasicStroke can set the stroke width and style.

- Basic Graphics Workflow:

JFrame - JPanel - Graphics/Graphics2D - Set color and stroke - Draw graphics

- Color in Java Graphics:

~ The Color class of java.awt package is used to define the color of graphics.

~ setColor() method is used to define the color before drawing.

Example: g2.setColor(Color.RED);

Predefined colors like RED, BLUE, GREEN, BLACK, YELLOW, etc can be used.

Custom colors can also be defined using RGB value.

- Object Field and Static Field:

Object Field: An object field, also known as an instance field, is a member of an individual object. Each object has its own copy of an object field.

Example:

```
class Human {  
    String name;  
    int age;  
}
```

Here name and age will be different for different objects of Human.

Static Field: A static field is a member of the class but not of an individual object. One single copy of the static field is there.

Example:

```
class Human {  
    String name;  
    static int population;  
}
```

Here name is object field whereas population is static field;
which is common to all objects of the class.

- Difference:

Object Field

- Is a member of individual object
- Separate copy for each object

Static Field

- Is a member of the class
- One common copy